Scenario 1: Employee Salary Analysis

Question:
Find the names and salaries of employees who earn more than the average salary in the company.

Answer:

- Use SELECT to display employee name and salary
- Use AVG() aggregate function to calculate the average salary
- Use a subquery to find the average salary
- Use WHERE clause to filter employees whose salary is greater than the average salary

```
SELECT name, salary
FROM Employee
WHERE salary > (
        SELECT AVG(salary)
        FROM Employee);
```

Scenario 2: Customer Orders without Matching Records

Question:
Retrieve a list of customer names who have not placed any orders.

Answer:

- Use SELECT to display customer names
- Use FROM to select the Customer table
- Use LEFT JOIN to compare customers with orders
- Use WHERE clause to filter customers who have no matching order records

```
SELECT c.customer_name
FROM Customer c
LEFT JOIN Orders o ON c.customer_id = o.customer_id
WHERE o.customer_id IS NULL;
```

Scenario 3: Product Sales Summary

Question:
Display the total sales amount for each product.

Answer:

- Use SELECT to display product_id (or product_name)
- Use SUM() aggregate function to calculate the total sales amount
- Use FROM to select the Sales table
- Use GROUP BY to calculate totals for each product separately

```
SELECT product_name,
    SUM(amount) AS Total_Sales
FROM Sales
GROUP BY product_name;
```

Scenario 4: Department-Wise Employee Count

Question:
List each department name with the number of employees working in it.

Answer:

- Use SELECT to display the department name
- Use COUNT() aggregate function to count employees in each department
- Use FROM to select employee table
- Use GROUP BY to group employees department-wise.

```
SELECT department_name,
        COUNT(employee_id) AS number_of_employees
FROM employee
```

GROUP BY department_name

Scenario 5: Top 3 Highest Sales

Question:
Find the top 3 highest sales transactions.

Answer:

- Use SELECT to display the required columns
- Use FROM to select the Sales table
- Use ORDER BY to sort sales amount in descending order
- Use LIMIT 3 to return only the highest 3 transactions

SELECT sales_id, amount

FROM Sales

ORDER BY amount DESC

LIMIT 3;

Scenario 6: Calculate Employee Salary Ranks by Department

Question:
Write a query to display each employee's name, department name, salary, and their salary rank within their respective department.

Answer

- Use SELECT to display employee name, department name, salary, and rank
- Use FROM to select the Employee table
- Use INNER JOIN to combine Employee and Department tables using department ID
- Use RANK() OVER(PARTITION BY … ORDER BY …) to calculate salary rank inside each department
- Use ORDER BY to display ranked results

SELECT e.employee_name, d.department_name, e.salary,

```
    RANK() OVER (PARTITION BY d.department_name ORDER BY e.salary DESC) AS
    salary_rank

FROM Employee e

INNER JOIN Department d ON e.department_id = d.department_id

ORDER BY d.department_name, salary_rank;
```
